Bernstein–Vazirani Algorithm

Nishchay Kumar
Roll No: 252IS024

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1. Problem Definition

Bernstein–Vazirani Algorithm finds the hidden binary string $s \in \{0,1\}^n$ in **one query** using superposition and interference.
Given black-box $f : \{0,1\}^n \to \{0,1\}$ with

$$f(x) = x \cdot s \mod 2 = x_1 s_1 \oplus x_2 s_2 \oplus \cdots \oplus x_n s_n$$

Goal: find hidden s.

$$s \in \{0,1\}^n$$

2. Optimal Classical Algorithm

Requires n **queries**:

$$f(00\ldots010\ldots00) = s_k$$

Example $s = 101$ $(n = 3)$:

$$f(100) = 1 \cdot 1 \oplus 0 \cdot 0 \oplus 0 \cdot 1 = 1 \qquad s_1 = 1$$
$$f(010) = 0 \cdot 1 \oplus 1 \cdot 0 \oplus 0 \cdot 1 = 0 \qquad s_2 = 0$$
$$f(001) = 0 \cdot 1 \oplus 0 \cdot 0 \oplus 1 \cdot 1 = 1 \qquad s_3 = 1$$

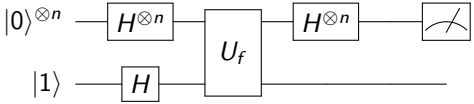
Time complexity: $O(n)$.

3. Quantum Algorithm

Key idea:

- Use superposition to evaluate all inputs at once
- Phase kickback encodes hidden string
- Hadamard extracts s

Quantum Circuit:



where $U_f|x\rangle|y\rangle = |x\rangle|y \oplus f(x)\rangle$.

Circuit Steps

- Initialize: $|0\rangle^{\otimes n} |1\rangle$
- Apply Hadamard gates
- Apply Oracle U_f
- Apply Hadamard again
- Measurement

Step-by-Step Explanation

Step 1: Initialization :

$$|0\rangle^{\otimes n}\ |1\rangle$$

Step 2: Apply Hadamard:

$$H^{\otimes n}|0\rangle^{\otimes n}\ H|1\rangle$$

$$= \frac{1}{\sqrt{2^n}} \sum_{x\in\{0,1\}^n} |x\rangle\ |-\rangle$$

Step 3: Apply Oracle U_f

Phase Kickback:

$$|x\rangle |-\rangle \xrightarrow{U_f} (-1)^{f(x)} |x\rangle |-\rangle$$

Now:

$$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} |x\rangle |-\rangle \xrightarrow{U_f} \frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle |-\rangle$$

Step 4: Apply Hadamard Again

$$\frac{1}{\sqrt{2^n}} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} |x\rangle |-\rangle \xrightarrow{H^{\otimes n}}$$

$$= \frac{1}{2^n} \sum_{x \in \{0,1\}^n} (-1)^{f(x)} \sum_{y \in \{0,1\}^n} (-1)^{x \cdot y} |y\rangle |-\rangle$$

$$= \frac{1}{2^n} \sum_{y \in \{0,1\}^n} \left[(-1)^{x \cdot s} (-1)^{x \cdot y} \right] |y\rangle |-\rangle$$

Step 5: Measurement

We have to find $|s\rangle_n$

So,

$$|y\rangle_n = |s\rangle_n$$

Now,

$$= \frac{1}{2^n} \sum_{y \in \{0,1\}^n} \left[(-1)^{s \cdot x} (-1)^{x \cdot y}\right] |y\rangle_n$$

$$= \frac{1}{2^n} \sum_{y \in \{0,1\}^n} \left[(-1)^{2(s \cdot x)}\right] |s\rangle_n$$

$$= \frac{1}{2^n} \cdot 2^n \, |s\rangle_n$$

$$= |s\rangle_n$$

Example: $n = 3$, $s = 101$

Step 1:

$$|000\rangle|1\rangle$$

Step 2:

$$H^{\otimes 3} \to \frac{1}{\sqrt{8}}(|000\rangle + |001\rangle + \cdots + |111\rangle)\,|-\rangle$$

Step 3: Apply Oracle U_f

$$|x\rangle \to (-1)^{f(x)}|x\rangle$$

$$f(x) = x \cdot s \pmod 2$$

x	$x \cdot s$ **(mod 2)**	$(-1)^{x \cdot s}$
000	0	$+1$
001	1	-1
010	0	$+1$
011	1	-1
100	1	-1
101	0	$+1$
110	1	-1
111	0	$+1$

State After Oracle

State after Oracle:

$$\frac{1}{\sqrt{8}}(|000\rangle - |001\rangle + |010\rangle - |011\rangle - |100\rangle + |101\rangle - |110\rangle + |111\rangle)$$

Apply Hadamard Again (Interference):

$$H^{\otimes n}|x\rangle = \frac{1}{2^n} \sum_{y \in \{0,1\}^n} [(-1)^{x \cdot s}(-1)^{x \cdot y}]\,|y\rangle$$

Case 1: $y = s$

$$\text{Amplitude}(y) = \frac{1}{8}\sum_{x=0}^{7}(-1)^{x\cdot s}(-1)^{x\cdot y}$$

$$(-1)^{x\cdot s}(-1)^{x\cdot y} = 1$$

$$\Rightarrow \frac{1}{8} \times 8 = 1$$

Conclusion:
All incorrect states cancel due to destructive interference.
Only the correct state remains with amplitude 1.

Case 2: $y \neq s$

$$\text{Amplitude}(y) = \frac{1}{8}\sum_{x=0}^{7}(-1)^{f(x)}(-1)^{x\cdot y}$$

$$(-1)^{f(x)}(-1)^{x\cdot y} = (-1)^{x\cdot s}(-1)^{x\cdot y} = (-1)^{x\cdot(s\oplus y)}$$

Let:

$$w = s \oplus y$$

Since $y \neq s \Rightarrow w \neq 0$

$$\sum_{x}(-1)^{x\cdot w} = 0$$

$$\Rightarrow \text{Amplitude}(y) = 0$$

Let:

$$s = 101, \quad y = 000$$

$$w = s \oplus y = 101$$

x	$x \cdot w$	$(-1)^{x \cdot w}$
000	0	$+1$
001	1	-1
010	0	$+1$
011	1	-1
100	1	-1
101	0	$+1$
110	1	-1
111	0	$+1$

$$(+1 - 1 + 1 - 1 - 1 + 1 - 1 + 1) = 0$$

$$A(y) = \frac{1}{8} \times 0 = 0$$

Conclusion:

When $y \neq s$, phase term produce equal positive and negative contributions , so they cancel out.

Classical vs Quantum

Classical	**Quantum (BV)**
n queries	**1 query**
Query basis states	Superposition of all 2^n
No parallelism	Massive parallelism + interference
$O(n)$ time	$O(1)$ queries

Perfect speed-up for this problem!

Reference

Thomas G. Wong, *Introduction to Classical and Quantum Computing*,